Useful Memory Has a Horizon

Temporal Validity, Roughness, and Structural Limits for Finite-Memory Tracking under Drift

Vinicius Parreira

Independent Researcher

figueiredo@pm.me

May 2026

Abstract

In stationary settings, memory is an unambiguous statistical resource: more data improves estimation because the target distribution does not move. Under drift, that premise fails. Retained evidence reduces sampling noise, but it also accumulates temporal misalignment with the present. Finite-memory tracking is therefore governed by a temporal-validity problem: how long can past evidence continue to improve estimation of the current distribution before staleness overtakes variance reduction?

We study this question through a carrier-roughness useful-memory horizon law. The main theorem object is an abstract upper law of the form

$$\mathbb{E}\, d\Big(\widehat{P}_t^{(n)}, P_t\Big) \le C_K n^{-a} + C_S \zeta n^H,$$

which couples a finite-sample carrier exponent a to a temporal roughness index $H \in (0,1]$. Optimizing the trade-off yields the useful-memory scale $n^*(a,H) \asymp (C_K/\zeta)^{1/(a+H)}$ and the corresponding optimized error law. The familiar cube-root horizon appears as the special case $a = 1/2$, $H = 1$.

The analysis anchors this law with a rigorous minimum-kernel result: in a one-dimensional bounded-support fixed-span triangular array, $\mathbb{E}W_2(\widehat{P}_n^{\mathrm{tri}}, \bar{P}_n) = O(n^{-1/2})$ via a quantile/Bahadur route. A Gaussian witness lower bound shows that on this main carrier slice, the useful-memory scale is structural rather than an artifact of a particular estimator. The analysis also identifies two broader settings: a useful layer in which low intrinsic dimension preserves the carrier exponent relative to the i.i.d. mixture benchmark, and a practical measurement layer based on fixed-ε Sinkhorn geometry whose carrier appears stable in finite-sample experiments.

The empirical section makes the theory visible through four signatures: a finite-memory U-curve, roughness-dependent horizon scaling, carrier inheritance beyond the minimum kernel, and detector-silent staleness. The resulting picture is that useful memory has a path-dependent horizon, and temporal validity can fail before changepoint evidence becomes statistically visible.

Keywords: tracking under drift; useful memory; temporal validity; carrier-roughness useful-memory horizon law; invalidity gap; Wasserstein distance.
MSC 2020: 93D05, 93D30, 68P99.

1 Introduction

In stationary statistics, memory is an unambiguous resource: larger samples reduce variance because the target distribution does not change. Under drift, that logic fails. In streaming estimation, monitoring, and online adaptation, the evidence retained in memory was generated by distributions that may no longer represent the present. More memory therefore does two things at once: it reduces finite-sample noise, and it increases the age of the evidence used to estimate the current state.

That tension creates a horizon problem. Short windows are fresh but noisy. Long windows are stable but stale. The relevant question is not only whether memory helps, but how long retained evidence can remain temporally valid before drift-induced misalignment overtakes the remaining variance gain. The paper studies that question in distributional geometry and treats the horizon of useful memory as the main statistical object.

The central contribution is a carrier-roughness useful-memory horizon law. The starting point is an abstract decomposition

$$\mathbb{E}\, d\Big(\widehat{P}_t^{(n)}, P_t\Big) \le C_K n^{-a} + C_S \zeta n^H,$$

where $a > 0$ is the finite-sample carrier exponent of the chosen measurement layer, $H \in (0, 1]$ is the temporal roughness exponent of the drift path, and ζ is the roughness amplitude. Optimizing this balance yields the useful-memory scale

$$n^*(a, H) \asymp (C_K/\zeta)^{1/(a+H)}.$$

The cube-root law is recovered as the special case $a = 1/2$, $H = 1$, corresponding to a root-n carrier and worst-case linear staleness. The paper is therefore not about the cube-root law alone. It is about the larger law that explains when cube-root scaling appears, when it does not, and why the horizon is path dependent.

To make this law concrete, the paper proceeds in three steps.

First, it identifies a *minimum kernel*: a one-dimensional bounded-support fixed-span triangular-array setting in which the carrier can be instantiated rigorously. In that regime, the quantile representation of W_2^2, a triangular-array Bahadur expansion, and a design-effect comparison yield the canonical carrier

$$\mathbb{E}\, W_2\Big(\widehat{P}_n^{\text{tri}}, \bar{P}_n\Big) = O(n^{-1/2}).$$

This supplies a concrete $a = 1/2$ slice of the abstract law.

Second, the paper identifies a *useful layer*: low-dimensional or low-intrinsic-dimensional regimes in which the triangular-array carrier tracks the i.i.d. mixture benchmark with only a constant-level gap. This extends the theory beyond the one-dimensional proof regime without pretending to solve the full high-dimensional raw-W_2 problem.

Third, it studies a *practical layer*: fixed-ε Sinkhorn geometry as a measurement layer whose carrier appears stable in finite samples even where raw Wasserstein geometry is dimensionally fragile. The point is not that this layer has already been proved in full generality, but that it provides a coherent measurement-layer regime for the same horizon law.

The theory is complemented by a structural lower bound. On the main carrier slice $a = 1/2$, a Gaussian witness construction shows that the useful-memory scale is not merely the optimum of one procedure. At the Lipschitz endpoint, the same cube-root scale appears as a subclass lower bound, making the horizon structural rather than estimator specific.

The empirical logic mirrors the theory. The paper first makes the basic variance–staleness trade-off visible through a finite-memory U-curve. It then shows roughness-dependent horizon scaling across path classes, carrier behavior beyond the minimum kernel, and a measurable invalidity gap between temporal validity and changepoint evidence. The central conceptual distinction is simple: temporal validity is not changepoint evidence. A detector can remain statistically silent while retained evidence is already stale for the present.

This framing is adjacent to several literatures—dynamic regret, adaptive windowing, locally stationary smoothing, adaptive filtering, and distributional tracking under transport geometry—but it is not reducible to any one of them. The question here is narrower and more structural: for finite-memory tracking under drift, how long can retained evidence continue to improve estimation of the present before age becomes the dominant source of error?

The paper proceeds as follows. Section 2 develops the carrier-roughness useful-memory horizon law, including the minimum-kernel result, the structural lower bound, and the broader useful and practical settings. Section 3 presents the corresponding finite-sample evidence. Section 4 discusses related work, and Section 5 discusses scope, limitations, and open problems.

2 The Carrier-Roughness Useful-Memory Horizon Law

Under drift, memory is both a statistical resource and a temporal liability. Retaining more data reduces finite-sample noise, but it also increases the age of the evidence used to represent the present. The object of interest is therefore not a detector threshold or a particular adaptive rule, but the error of finite-memory representation itself.

For a finite-memory estimator with lag weights $w_0, \ldots, w_{n-1}$, the natural age variable is the effective lag

$$A(w) \coloneqq \sum_{j=0}^{n-1} j\, w_j .$$

Its corresponding staleness at time t is the weighted transport misalignment

$$S_t(w) \coloneqq \sum_{j=0}^{n-1} w_j\, W_2(P_{t-j}, P_t)\,.$$

For a uniform window, $A(w) = (n-1)/2$, so window length and temporal age are the same object up to constants. This is why memory length becomes a statistical question rather than a purely algorithmic one.

2.1 Abstract Upper Law

The top-level theorem separates finite-sample carrier behavior from drift-induced staleness.

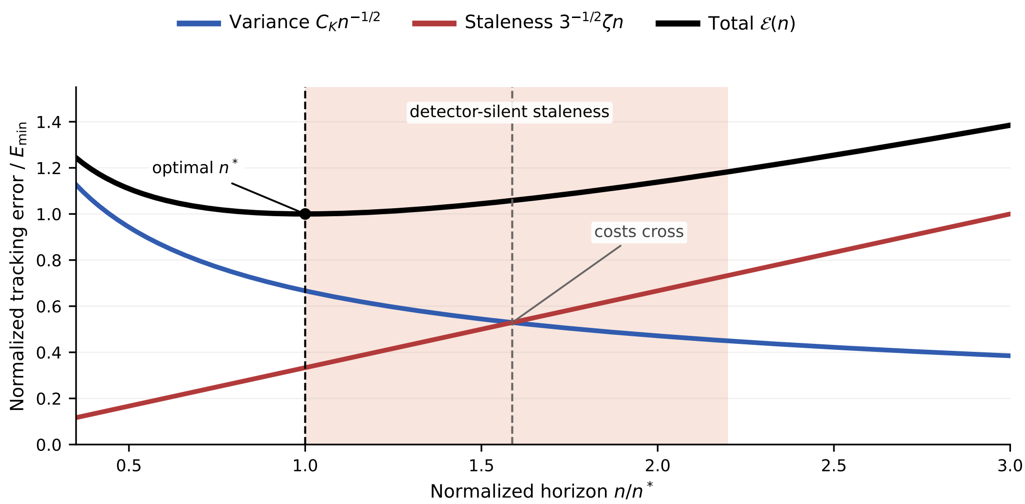

Figure 1: Two clocks of drift. The finite-sample term falls with horizon, the staleness term rises with age, and total error is minimized at the useful-memory horizon n^*. The shaded region represents detector-silent staleness: retained evidence can lose temporal validity before enough changepoint evidence accumulates to trigger an alarm.

Theorem 2.1 (Abstract upper law)**.** *Let d be a probability metric satisfying the triangle inequality. At time t, let P_t be the present target law, let $\bar{P}_t^{(n)}$ be the window target, and let $\widehat{P}_t^{(n)}$ be the estimator built from the last n observations. Assume that for some constants $C_K > 0$, $C_S > 0$, exponent $a > 0$, roughness index $H \in (0, 1]$, and amplitude $\zeta > 0$,*

$$\mathbb{E}\, d\Big(\widehat{P}_t^{(n)}, \bar{P}_t^{(n)}\Big) \le C_K n^{-a}$$

and

$$d\Big(\bar{P}_t^{(n)}, P_t\Big) \le C_S \zeta n^H$$

hold for all n in the regime of interest. Then

$$\mathbb{E}\, d\Big(\widehat{P}_t^{(n)}, P_t\Big) \le C_K n^{-a} + C_S \zeta n^H.$$

Proof. By the triangle inequality,

$$d\Big(\widehat{P}_t^{(n)}, P_t\Big) \le d\Big(\widehat{P}_t^{(n)}, \bar{P}_t^{(n)}\Big) + d\Big(\bar{P}_t^{(n)}, P_t\Big).$$

Taking expectations and applying the two assumptions yields the claim. □

The proof is immediate, but the theorem makes the modular structure explicit: the path class supplies the staleness term, while the measurement layer supplies the carrier term.

Corollary 2.2 (Optimized useful-memory horizon law)**.** *Under the assumptions of theorem 2.1, the upper envelope*

$$\Phi(n) := C_K n^{-a} + C_S \zeta n^H$$

is minimized, up to constant factors, at

$$n^*(a,H) \asymp (C_K/\zeta)^{1/(a+H)},$$

and the corresponding optimized error scale is

$$\Phi(n^*) \asymp C_K^{H/(a+H)}\zeta^{a/(a+H)}.$$

Proof. Balance the decreasing carrier term against the increasing staleness term:

$$C_K n^{-a} \asymp C_S \zeta n^H.$$

This gives $n^{a+H} \asymp C_K/\zeta$, hence the displayed horizon law. Substituting that scale back into either term gives the optimized error scale. □

The cube-root law is recovered when $a = 1/2$ and $H = 1$. More generally, the carrier exponent and the roughness exponent jointly determine the useful-memory scale. There is no universal horizon independent of those two ingredients.

2.2 Path Classes and the Roughness Index

The staleness side of the law is controlled by a deterministic path class. For $H \in (0,1]$ and $\zeta > 0$, write

$$\mathfrak{P}_H(\zeta) := \left\{(P_s)_{s\in\mathbb{Z}} : W_2(P_{t-j}, P_t) \le \zeta j^H \text{ for all } j \ge 0\right\}.$$

The case $H = 1$ is the local Lipschitz envelope; smaller H correspond to slower accumulation of staleness with lag.

Lemma 2.3 (Holder staleness bound via averaged couplings)**.** *Let*

$$\bar{P}_t^{(n)} := \frac{1}{n}\sum_{j=0}^{n-1} P_{t-j}.$$

If the target path belongs to $\mathfrak{P}_H(\zeta)$, then

$$W_2^2\left(\bar{P}_t^{(n)}, P_t\right) \le \frac{\zeta^2}{n}\sum_{j=0}^{n-1} j^{2H},$$

and therefore

$$W_2\left(\bar{P}_t^{(n)}, P_t\right) \le c_H \zeta n^H$$

for a constant $c_H > 0$ depending only on H.

Proof. For each lag j, let π_j be an optimal coupling between P_{t-j} and P_t. Their average

$$\bar{\pi} := \frac{1}{n}\sum_{j=0}^{n-1} \pi_j$$

couples $\bar{P}_t^{(n)}$ with P_t, so

$$W_2^2\left(\bar{P}_t^{(n)}, P_t\right) \le \int \|x-y\|^2 \, d\bar{\pi}(x,y) = \frac{1}{n}\sum_{j=0}^{n-1} W_2^2(P_{t-j}, P_t).$$

Applying the Holder envelope and taking square roots gives the result. □

Combining lemma 2.3 with corollary 2.2 yields the family

$$n^*(a, H) \asymp (C_K/\zeta)^{1/(a+H)}.$$

On the canonical statistical slice $a = 1/2$,

$$n^*(1/2, H) \asymp (C_K/\zeta)^{2/(1+2H)}.$$

At $H = 1$, this recovers the cube-root horizon; at $H = 1/2$, it yields the intermediate $n^* \asymp C_K/\zeta$ regime. These are members of the same law, not separate theorem stories.

2.3 Minimum Kernel: A Rigorous Carrier Instantiation

The abstract law needs at least one rigorous carrier instantiation. The minimum kernel is the bounded-support fixed-span one-dimensional triangular array. It provides the first clean carrier feeding the abstract horizon law.

Proposition 2.4 (Minimum-kernel carrier)**.** *Let $X_{n,1}, \dots, X_{n,n}$ be independent one-dimensional observations with laws $P_{n,1}, \dots, P_{n,n}$, and define the window mixture*

$$\bar{P}_n = \frac{1}{n} \sum_{j=1}^{n} P_{n,j}.$$

Assume:

(i) all $P_{n,j}$ are supported on a common compact interval $[L, U]$;

(ii) the within-window drift span is uniformly bounded in n;

(iii) on an interior quantile band $u \in [\varepsilon, 1-\varepsilon]$, the mixture density $\bar{f}_n$ is bounded below and uniformly Holder;

(iv) the triangular-array empirical quantile process admits a Bahadur representation with integrated remainder $o(n^{-1})$ on that interior band.

Then

$$\mathbb{E}\, W_2^2\left(\widehat{P}_n^{\mathrm{tri}}, \bar{P}_n\right) = O(n^{-1}),$$

and therefore

$$\mathbb{E}\, W_2\left(\widehat{P}_n^{\mathrm{tri}}, \bar{P}_n\right) = O(n^{-1/2}).$$

Proof sketch. In one dimension,

$$W_2^2\left(\widehat{P}_n^{\mathrm{tri}}, \bar{P}_n\right) = \int_0^1 \left(\widehat{q}_n(u) - q_n(u)\right)^2 du,$$

so the problem reduces to the integrated L_2 error of the empirical quantile process. On the interior band, the Bahadur representation gives

$$\widehat{q}_n(u) - q_n(u) = \frac{u - \widehat{F}_n(q_n(u))}{\bar{f}_n(q_n(u))} + R_n(u).$$

The leading term has integrated variance $O(n^{-1})$ because the observations are independent and the fixed-span triangular design changes only constants relative to the i.i.d. mixture benchmark. The remainder splits into an empirical increment term and a Taylor term; the empirical increment term is the dominant one, but under the stated interior regularity it still satisfies an integrated $o(n^{-1})$ bound. Bounded support makes the boundary-band contribution lower-order. Summing the leading term, the lower-order cross term, and the integrated remainder gives $\mathbb{E}W_2^2 = O(n^{-1})$, and Jensen yields the displayed W_2 rate.

The main point is the exponent. Inside this safe zone, the triangular-array carrier remains root-n and differs from the i.i.d. mixture benchmark only through a design effect.

2.4 A Structural Lower Bound

The upper law would be much less interesting if it were only the optimum of one estimator. The lower bound shows that, on the main carrier slice $a = 1/2$, the useful-memory scale is imposed by the drift geometry itself.

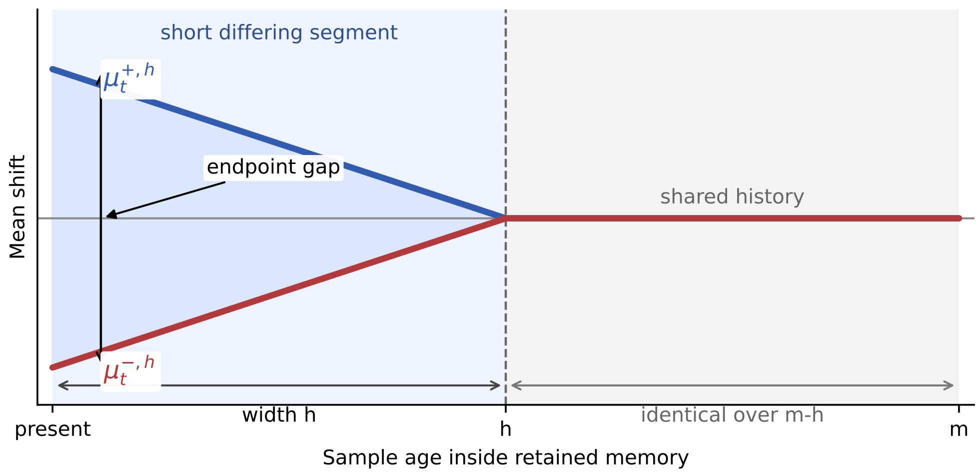

Figure 2: Gaussian witness for the structural lower bound. The two paths share most of the retained history and differ only on a recent segment of width h, yet remain separated at the present. Optimizing that recent segment already produces the same horizon exponent as the upper law on the main carrier slice.

Proposition 2.5 (Structural lower bound on the main carrier slice). *Fix $H \in (0, 1]$ and work on the Gaussian location subclass with known variance σ^2 and witness paths*

$$\mu_{t-j}^{\pm,h} = \pm\beta(h-j)_+^H, \qquad j = 0, 1, \ldots, h,$$

where $\beta \le \zeta$ enforces the roughness budget. Then there exists a constant $c_H > 0$ depending only on H such that the endpoint tracking risk over this subclass satisfies

$$\inf_{\widehat{P}} \sup_{P \in \mathcal{G}_{H,\zeta}} \mathbb{E}\, W_2\Big(\widehat{P}_t, P_t\Big) \ge c_H\, \sigma^{2H/(2H+1)} \zeta^{1/(2H+1)}.$$

Consequently, on the main carrier slice $a = 1/2$, the useful-memory scale

$$h^*(H) \asymp (\sigma/\zeta)^{2/(2H+1)}$$

is structural even on this restricted subclass.

Proof sketch. Compare two hypotheses generated by the witness paths $\mu_{t-j}^{+,h}$ and $\mu_{t-j}^{-,h}$. At the present time, the endpoint separation is of order βh^H. On the Gaussian location subclass, this is also the W_2 separation. The Kullback–Leibler divergence between the two hypotheses is proportional to

$$\beta^2 \sum_{r=1}^{h} r^{2H} / \sigma^2.$$

Choosing

$$\beta = \min\left\{ \zeta, \frac{\sigma}{2\sqrt{\sum_{r=1}^{h} r^{2H}}} \right\}$$

keeps the testing problem in the Le Cam regime while respecting the roughness budget. The resulting two-point lower bound is of order βh^H. Balancing the active roughness constraint against the testing constraint gives $h^*(H) \asymp (\sigma/\zeta)^{2/(2H+1)}$ and therefore the displayed lower law.

At $H = 1$, this recovers the cube-root horizon and the lower scale $\sigma^{2/3}\zeta^{1/3}$. The lower bound is subclass based rather than class-tight, but that is enough for the paper's central claim: the horizon is not just an estimator artifact.

2.5 Beyond the Minimum Kernel

The one-dimensional minimum kernel is the first rigorous carrier instantiation. Two broader settings are also important for interpreting the law.

Useful layer. In low-dimensional or low-intrinsic-dimensional settings, the natural question is whether the triangular-array carrier inherits the same exponent as the i.i.d. mixture benchmark up to a constant-level gap. The clearest evidence comes from embedded $k = 1$ support inside a larger ambient space, where triangular and i.i.d. slopes remain near the $a = 1/2$ slice under fixed span.

Practical layer. For a practical measurement layer such as fixed-ε debiased Sinkhorn, the corresponding question is whether the induced finite-sample carrier is stable enough to feed the same horizon law. The experiments reported below are consistent with that interpretation: across the tested regularization values, they show a stable measurement-layer signal rather than a qualitative change of behavior.

Together, these settings show how the abstract law extends beyond the one-dimensional minimum kernel while keeping the formal claims within their proper scope.

Corollary 2.6 (EWMA analogue)**.** *For exponential weights* $w_j = \alpha(1-\alpha)^j$*, the effective lag is* $\tau_\alpha = (1-\alpha)/\alpha$ *and the effective sample size scales as* $n_{\text{eff}} \asymp 1/\alpha$*. Balancing the same carrier and staleness terms therefore yields the same exponent in the effective-memory scale. In particular, on the main slice* $a = 1/2$*,* $H = 1$*, the EWMA analogue of the cube-root horizon is recovered up to constants.*

3 Empirical Evidence

The empirical section makes the theoretical picture visible in finite samples. Its role is twofold: to show the main signatures of the variance–staleness law, and to calibrate the broader useful and practical settings beyond the minimum-kernel result.

Four signatures matter. First, horizon sweeps should exhibit a finite optimum rather than a boundary solution. Second, that optimum should move with temporal roughness. Third, carrier inheritance should remain visible beyond the minimum kernel in useful and practical regimes. Fourth, temporal validity can fail before changepoint evidence becomes statistically visible.

3.1 A Visible Finite-Memory Optimum

The Gaussian horizon sweep isolates the variance–staleness trade-off directly. Small windows are variance dominated, large windows are stale, and the minimum lies at an interior horizon rather than at either boundary.

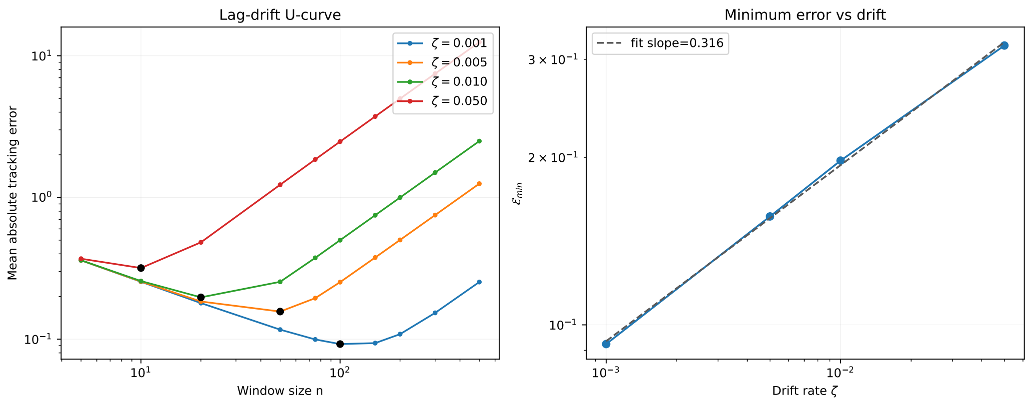

Figure 3: Structural U-curve for finite-memory tracking. Tracking error first falls with horizon and then rises once staleness overtakes the remaining variance gain. The optimum remains finite across drift strengths.

This is the most basic finite-memory signature. Under stationarity, more memory would continue to help. Under drift, the same horizon that reduces variance also ages the evidence.

3.2 Roughness-Dependent Horizon Scaling

The next experiment varies the roughness index directly. For each Holder-type regime, the surrogate error obeys the same structural decomposition as the theory,

$$\mathcal{E}(n) \approx \sigma n^{-1/2} + \zeta n^{H},$$

and the experiment records the empirically optimal horizon across a sweep of ζ values.

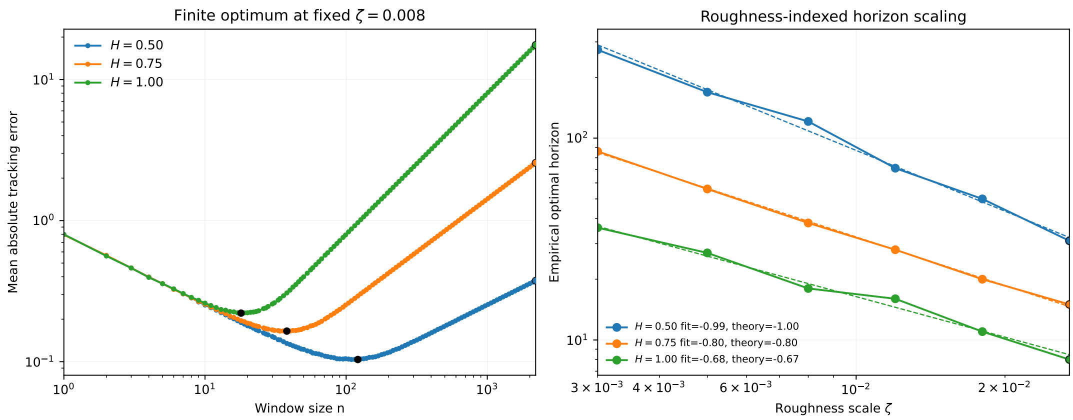

Figure 4: Roughness-dependent horizon scaling. Each path class still exhibits a finite optimum, but the location of that optimum changes with H. The worst-case Lipschitz cube-root law appears as the $H = 1$ member of a broader carrier-roughness useful-memory horizon family.

The point is structural rather than cosmetic. If the cube-root law were the only story, the right panel would collapse to a single slope. It does not. The empirical optima separate by roughness class, matching the path-dependent horizon logic of corollary 2.2.

3.3 Useful and Practical Carrier Regimes

The next figure summarizes two broader carrier settings: a useful setting based on low intrinsic dimension, and a practical measurement setting based on fixed- ε Sinkhorn geometry.

The left panel concerns bounded-support fixed-span samples with low intrinsic dimension. In that setting, the triangular-array carrier tracks the i.i.d. mixture benchmark rather than exhibiting a visibly different exponent.

The right panel concerns fixed-ε Sinkhorn geometry. It shows that the corresponding measurement-layer carrier remains stable across the tested regularization values and can therefore be read against the same abstract horizon law. Appendix A reports the corresponding summary table.

3.4 Detector-Silent Staleness

Temporal validity and changepoint evidence are different clocks. To make that separation explicit, consider a stream whose local drift rate ramps upward over

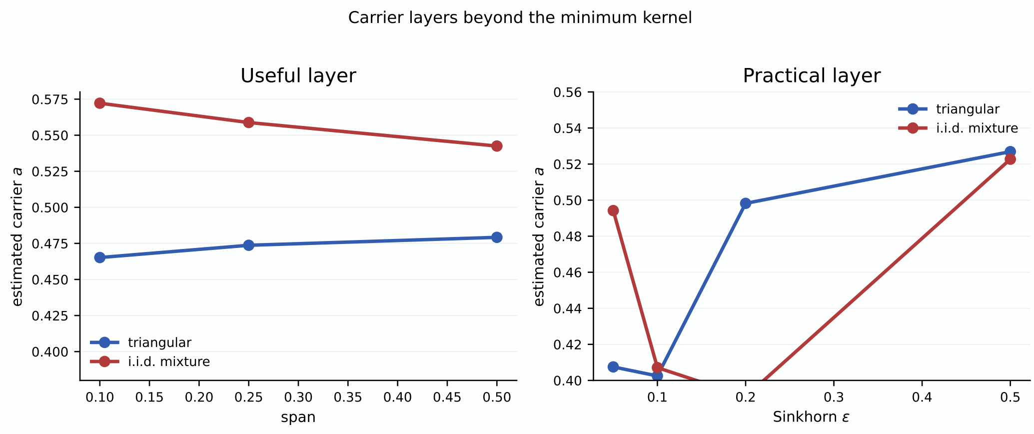

Figure 5: Useful and practical carrier settings. Left: in the useful setting, embedded low-intrinsic-dimensional support preserves a carrier near the $a = 1/2$ slice, and triangular slopes stay close to the i.i.d. mixture benchmark. Right: in the practical setting, fixed-ε Sinkhorn produces a measurement-layer carrier whose triangular and i.i.d. slopes remain qualitatively close across the tested ε values.

time. The useful-memory horizon n_t^* contracts as the ramp steepens, while a long operating horizon remains fixed. Define

$$\Delta_{\text{inv}} := t_{\text{detect}} - t_{\text{valid}},$$

where t_{valid} is the first time the operating horizon persistently exceeds the useful-memory scale, and t_{detect} is the first detector alarm that follows.

Across the reported runs, the measured gap remains positive. This is the empirical form of detector-silent staleness: retained evidence can already be invalid for the present while detector statistics remain below alarm threshold.

Table 1: Detector-threshold ablation for the invalidity gap. Across the tested ADWIN thresholds, mean detection time remains strictly later than mean validity-expiry time, so the measured invalidity gap stays positive throughout.

δ	$\bar{t}_{\text{valid}}$	$\bar{t}_{\text{detect}}$	$\overline{\Delta}_{\text{inv}}$
0.0005	1108.0	1671.0	563.0
0.0010	1108.0	1657.7	549.7
0.0020	1108.0	1641.7	533.7
0.0040	1108.0	1631.0	523.0

3.5 Horizon Misalignment Cost

The final empirical question is how sharply performance degrades when the operating horizon misses the useful-memory scale. Using the same roughness- indexed surrogate, fig. 7 plots excess error against the relative horizon gap $|\log(n/n^*)|$.

This is the operational reason the horizon matters. The optimum is not only finite; it is a scale around which performance is visibly better than on either side.

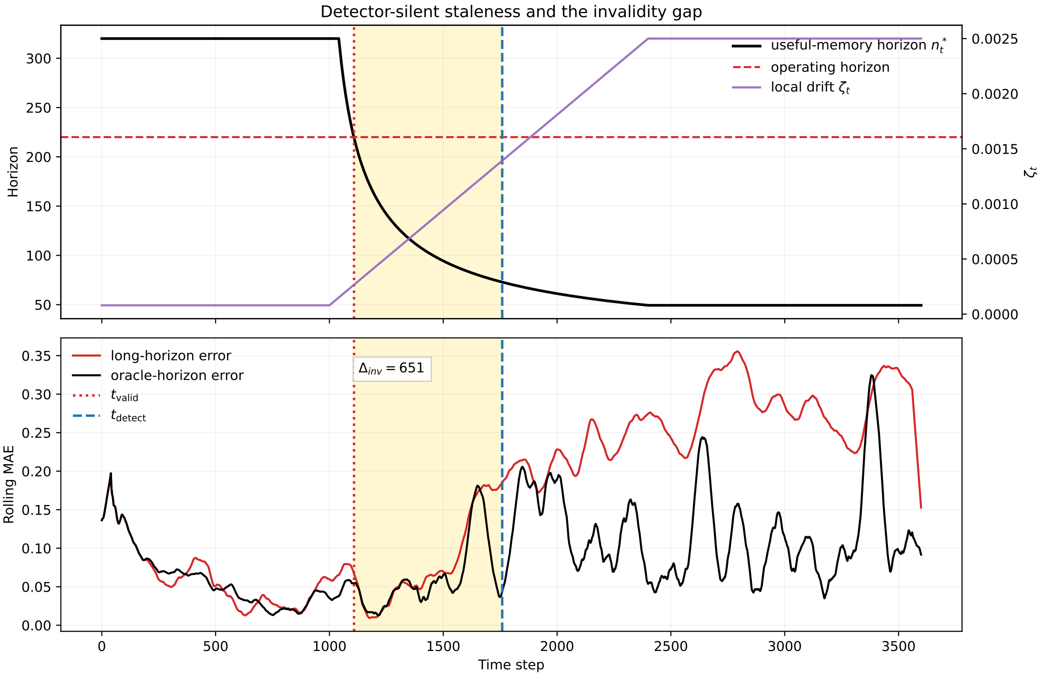

Figure 6: Detector-silent staleness. As drift intensifies, the useful-memory horizon contracts until the operating horizon becomes stale at t_{valid}; the detector responds only later at t_{detect}, leaving a strictly positive invalidity gap. The lower panel shows excess tracking error accumulating before the alarm appears.

Table 2: Detector-family comparison for the invalidity gap. Using the same drift ramp and operating horizon, the positive gap persists for both ADWIN and Page–Hinkley, even though their changepoint clocks differ.

Detector	$\bar{t}_{\text{valid}}$	$\bar{t}_{\text{detect}}$	$\overline{\Delta}_{\text{inv}}$
ADWIN	1108.0	1641.7	533.7
Page–Hinkley	1108.0	1410.6	302.6

3.6 What the Evidence Shows

Taken together, the experiments support four claims that match the theory. Finite-memory tracking exhibits a structural U-curve rather than an ever-growing memory benefit. The useful-memory scale is roughness dependent, with the cube-root law appearing as the $H = 1$ member of a broader family. Carrier stability remains visible beyond the minimum kernel in both useful and practical settings. Temporal validity can also fail before changepoint evidence appears, producing a measurable invalidity gap.

4 Related Work

Nonstationary optimization and dynamic regret. Dynamic-regret and variation-budget results show that under drift, window size or forgetting scale must depend on how quickly the target moves [Besbes et al., 2015, Cheung et al.,

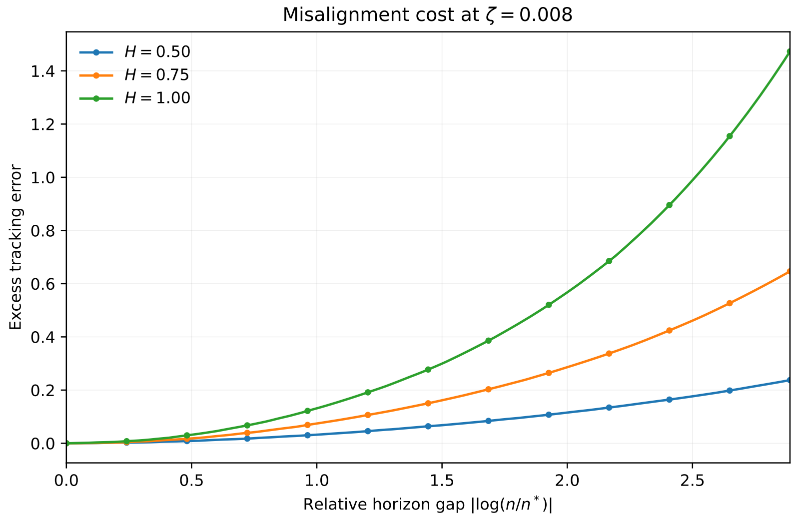

Figure 7: Cost of horizon misalignment. Excess tracking error is smallest near the useful-memory scale and rises once the operating horizon moves away from it. Being too short is noisy; being too long is stale.

2019, Yuan and Lamperski, 2020, Mazzetto and Upfal, 2023, Wang et al., 2024]. Those works are close in spirit, but their primary object is regret under a moving comparator. Here the primary object is different: the temporal validity of retained evidence in a distribution-tracking problem. The horizon is not a tuning parameter attached to regret analysis; it is the statistical object being characterized.

Adaptive windowing and drift detection. Adaptive-window and concept-drift methods ask when accumulated evidence is sufficient to react [Bifet and Gavaldà, 2007, Gama et al., 2004, 2014]. Hinder et al. [2023] emphasize that the relevant window scale is itself part of the problem. The present question is adjacent but distinct. Temporal validity is not changepoint evidence: memory can stop representing the present before the available data support a detector alarm. The invalidity gap makes that distinction measurable rather than rhetorical.

Adaptive filtering and state-space tracking. Classical adaptive filtering and state-space tracking also balance smoothing against responsiveness by tuning forgetting factors, gain schedules, or process noise models [Kalman, 1960, Jazwinski, 1970]. Those analyses usually work under stochastic state-space assumptions and ask for Bayes-optimal tracking of a latent process. The present paper instead studies deterministic Wasserstein drift envelopes and characterizes a worst-case temporal-validity horizon.

Locally stationary smoothing. Locally stationary time-series analysis likewise studies how window or bandwidth choice must adapt to time variation [Dahlhaus, 1997]. That literature is close in its bias–variance logic, but the moving object there is typically a local spectrum or regression function. Here the moving object is a path of distributions, and the age penalty is measured directly in transport geometry.

Distribution tracking under transport geometry. Wasserstein distance and Sinkhorn-type estimators are natural when drift lives on paths of distributions rather than on scalar losses [Fournier and Guillin, 2015, Genevay et al., 2019, Niles-Weed and Bach, 2019, Goldfeld et al., 2024, Akyildiz et al., 2024]. They supply the geometric language in which staleness becomes a transport error between past and present. In this setting, the trade-off is not simply between adaptation and variance, but between finite-sample accuracy and temporal misalignment measured in distributional geometry.

Wasserstein nonstationarity and robust optimization. Jiang, Li, and Zhang study online stochastic optimization under a Wasserstein nonstationarity measure and obtain distribution-aware regret guarantees [Jiang et al., 2020]. Keehan, Anderson, and Wiesemann analyze related drift-aware weighting questions in Wasserstein distributionally robust optimization [Keehan et al., 2025]. Amini and Vinas give a recent moment-method route to triangular-array concentration in W_1 [Amini and Vinas, 2026]. Those results are adjacent to the carrier-regime issue studied here, but they do not by themselves close the present W_2 window-mixture concentration problem. The present focus is narrower and more structural: for finite-memory tracking, how long can retained evidence continue to improve estimation of the current distribution before age becomes a dominant error source?

Freshness and Age of Information. The Age-of-Information literature treats age as a resource constraint and asks how update policies balance communication cost against stale information [Kaul et al., 2012, Yates et al., 2021]. The present paper gives a distributional analogue. Age is costly because it induces geometric error under drift, and temporal roughness determines how quickly that cost accumulates.

Across these neighboring literatures, the closest theme is that nonstationarity makes retention scale matter. The distinguishing claim here is more specific: finite memory obeys a carrier-roughness useful-memory horizon law, the worst-case cube-root law is the $a = 1/2$, $H = 1$ member of that family, and temporal validity can fail before change becomes statistically detectable.

5 Limitations and Scope

The central results of the paper are the carrier-roughness useful-memory horizon law, the minimum-kernel carrier on the canonical $a = 1/2$ slice, and the structural lower bound on that same slice. Broader extensions are stated more cautiously.

Carrier scope. The theory is modular in the carrier exponent a, but the main rigorous slice is the root-n carrier. The minimum kernel establishes that slice in one dimension under bounded support and fixed span. Outside that setting, the same balancing logic continues to hold, but the carrier must come from a different theorem. In particular, the paper does not claim a general raw-W_2 triangular-array carrier theorem in arbitrary dimension.

Useful and practical regimes. The useful setting and the fixed-ε Sinkhorn setting are supported here primarily through finite-sample evidence. The low-intrinsic-dimensional results indicate that the triangular-array carrier can track the i.i.d. mixture benchmark, and the Sinkhorn experiments indicate a stable measurement-layer signature across the tested regularization values. A theorem of the same strength as the minimum-kernel result is not established here for either setting.

Lower-bound scope. The lower bound is subclass based. The Gaussian witness is enough to show that on the main carrier slice $a = 1/2$, the useful-memory scale is structural rather than an artifact of a particular estimator. What remains open is class-tightness and constant-sharpness for full deterministic Holder path classes, especially away from the main slice and beyond the present witness geometry.

Weights and memory shape. The main text emphasizes uniform windows, with an EWMA corollary showing that the same exponent logic survives after effective-lag reparameterization. The paper does not investigate whether more general non-uniform weights can improve constants or alter exponents across broader path classes.

Empirical scope. The experiments are synthetic and structurally motivated. Their role is to make the theory visible and to calibrate broader carrier settings, not to demonstrate competitive performance on a benchmark suite. Real-data validation and direct comparison with adaptive horizon-selection rules remain outside the present scope.

Detectability versus validity. Detector-silent staleness is an empirical consequence of the structural theory, not the primary theorem object. The paper shows that temporal validity can fail before detector alarms appear, but it does not prove a universal detection-delay theorem across detector classes.

Open problems. The next technical questions are clear. The first is to establish carrier inheritance beyond the minimum kernel under natural low-dimensional conditions. The second is to understand when practical measurement layers, such as fixed- ε Sinkhorn, admit theorem-level carrier bounds. The third is to extend the lower bound beyond the main $a = 1/2$ slice with sharper constants and broader path classes. After that come non-uniform weighting, real-data validation, and stronger stochastic-process formulations.

6 Conclusion

Under drift, memory is never purely beneficial. It reduces finite-sample noise, but it also ages. The resulting trade-off makes useful memory a finite-horizon object rather than an unlimited statistical resource.

The paper formalizes that idea through a carrier-roughness useful-memory horizon law. The main theorem object is the abstract decomposition

$$\mathbb{E}\, d\Big(\widehat{P}_t^{(n)}, P_t\Big) \le C_K n^{-a} + C_S \zeta n^H,$$

whose optimized scale is

$$n^*(a,H) \asymp (C_K/\zeta)^{1/(a+H)}.$$

The cube-root horizon is therefore one member of a broader law: the special case $a = 1/2$, $H = 1$ corresponding to a root-n carrier under worst-case linear staleness.

The analysis anchors this law in a rigorous minimum-kernel result and a structural lower bound on the same main carrier slice. Together they show that, in the bounded-support fixed-span one-dimensional setting, the horizon is neither an artifact of a particular estimator nor a purely heuristic balance. The broader useful and practical settings show how the same law remains relevant beyond that initial result.

The empirical section makes the theory visible. It shows a structural U-curve, roughness-dependent horizon scaling, carrier behavior beyond the minimum kernel, and a strictly positive invalidity gap between temporal validity and changepoint evidence. The resulting message is both theoretical and operational: useful memory has a path-dependent horizon, and retained evidence can become invalid before change becomes statistically detectable.

The main open questions concern carrier inheritance beyond the minimum kernel, theorem-level measurement-layer results, and broader lower-bound extensions beyond the main $a = 1/2$ slice. These are natural extensions of the present theory, not changes to its central object.

Acknowledgements

With gratitude to everyone who contributed ideas, feedback, tools, encouragement, and care along the way. This work stands on the generosity of colleagues, collaborators, open-source maintainers, and the broader research communities that made it possible.

A Finite-Sample and Carrier-Layer Diagnostics

This appendix collects the numerical diagnostics that support the carrier side of the theory. These checks are not substitutes for the theorem statements. Their role is to show that the minimum-kernel, useful-regime, and practical-regime interpretations are consistent with the finite-sample behavior of the current experiment families.

A.1 Finite-Sample Geometry Check

For the finite-sample term, we estimate empirical W_2 between two i.i.d. samples from the same distribution using exact assignment in low and moderate dimension, then fit a log–log slope in n. We also probe the non-identically distributed fixed-span window directly in one dimension by comparing a triangular-array sample, with one draw from each drifted component, against an i.i.d. sample from the same window mixture. In that check the within-window span ζn^H is held fixed, so the experiment targets the minimum-kernel carrier interpretation rather than a large-span regime in which heterogeneity itself can become dominant. Finally, we check the Sinkhorn proxy at a few fixed values of ε.

Table 3: Empirical scaling check for the finite-sample term. The bounded-support i.i.d. W_2 experiment is close to the expected $n^{-1/2}$ rate in low dimension, the corresponding triangular-array check compares one sample from each drifted component against an i.i.d. sample from the window mixture, and the Sinkhorn proxy is reported at fixed ε. These rows are numerical support for the carrier-regime discussion rather than theorem statements.

Setting	Estimated slope	Comment
W_2, $d = 1$	-0.48	close to $-1/2$
W_2, $d = 2$	-0.41	slower decay
W_2, $d = 4$	-0.25	slower decay
W_2, $d = 5$	-0.24	slower decay
W_2, i.i.d. mixture, $d = 1$, $H = 0.5$	-0.50	fixed span $\zeta n^H = 0.25$
W_2, triangular, $d = 1$, $H = 0.5$	-0.45	fixed span $\zeta n^H = 0.25$
W_2, i.i.d. mixture, $d = 1$, $H = 1.0$	-0.47	fixed span $\zeta n^H = 0.25$
W_2, triangular, $d = 1$, $H = 1.0$	-0.41	fixed span $\zeta n^H = 0.25$
Sinkhorn $\varepsilon = 0.50$	-0.21	proxy constant changes
Sinkhorn $\varepsilon = 0.10$	-0.53	proxy constant changes
Sinkhorn $\varepsilon = 0.05$	-0.67	proxy constant changes

This table supports the carrier interpretation used in the main text. In low dimension, the raw i.i.d. Wasserstein term is compatible with a root-n carrier, whereas higher-dimensional raw Wasserstein slows down. In the one-dimensional fixed-span triangular-array check, the recovered slopes stay near the corresponding i.i.d. mixture baseline, which is consistent with the minimum-kernel and design-effect reading used in proposition 2.4.

A.2 Useful and Practical Carrier Regimes

The next table summarizes the bridge regimes beyond the minimum kernel. The useful rows compare triangular and i.i.d. mixture carriers on embedded low-intrinsic-dimensional support across span. The practical rows compare fixed-ε Sinkhorn carriers across regularization values.

These rows should be read as calibration rather than proof. The useful-regime rows support the claim that low intrinsic dimension preserves a carrier near the canonical $a = 1/2$ slice without a dramatic triangular-versus-benchmark gap. The practical-regime rows support the claim that fixed-ε Sinkhorn acts as a stable measurement layer in the current synthetic setting, even though the paper does not claim a general theorem for that layer.

Table 4: Summary rows for the useful and practical carrier layers. The useful slice compares triangular and i.i.d. mixture carriers across span, while the practical slice compares fixed-ϵ Sinkhorn across ϵ.

Layer	Setting	Parameter	Carrier a
useful	triangular ambient d=8, intrinsic k=1, span=0.10	0.10	0.4652
useful	iid mixture ambient d=8, intrinsic k=1, span=0.10	0.10	0.5722
useful	triangular ambient d=8, intrinsic k=1, span=0.25	0.25	0.4737
useful	iid mixture ambient d=8, intrinsic k=1, span=0.25	0.25	0.5588
useful	triangular ambient d=8, intrinsic k=1, span=0.50	0.50	0.4792
useful	iid mixture ambient d=8, intrinsic k=1, span=0.50	0.50	0.5425
practical	Sinkhorn iid mixture eps=0.50	0.50	0.5227
practical	Sinkhorn triangular eps=0.50	0.50	0.5269
practical	Sinkhorn iid mixture eps=0.20	0.20	0.3910
practical	Sinkhorn triangular eps=0.20	0.20	0.4982
practical	Sinkhorn iid mixture eps=0.10	0.10	0.4071
practical	Sinkhorn triangular eps=0.10	0.10	0.4025
practical	Sinkhorn iid mixture eps=0.05	0.05	0.4942
practical	Sinkhorn triangular eps=0.05	0.05	0.4075

References

O. Deniz Akyildiz, Pierre Del Moral, and Joaquín Miguez. Gaussian entropic optimal transport: Schr"odinger bridges and the sinkhorn algorithm. *CoRR*, abs/2412.18432, 2024. doi: 10.48550/arXiv.2412.18432.

Arash A. Amini and Luciano Vinas. Wasserstein concentration of empirical measures for dependent data via the method of moments. *CoRR*, abs/2601.07228, 2026. doi: 10.48550/arXiv.2601.07228.

Omar Besbes, Yonatan Gur, and Assaf Zeevi. Non-stationary stochastic optimization. *Operations Research*, 63(5):1227–1244, 2015. doi: 10.1287/OPRE.2015.1408.

Albert Bifet and Ricard Gavaldà. Learning from time-changing data with adaptive windowing. In *Proceedings of the 2007 SIAM International Conference on Data Mining (SDM)*, pages 443–448, 2007. doi: 10.1137/1.9781611972771.42.

Wang Chi Cheung, David Simchi-Levi, and Ruihao Zhu. Learning to optimize under non-stationarity. In *Proceedings of the Twenty-Second International Conference on Artificial Intelligence and Statistics*, volume 89 of *Proceedings of Machine Learning Research*, pages 1079–1087. PMLR, 2019. URL https://proceedings.mlr.press/v89/cheung19b.html.

Rainer Dahlhaus. Fitting time series models to nonstationary processes. *The Annals of Statistics*, 25(1):1–37, 1997. doi: 10.1214/aos/1034276620.

Nicolas Fournier and Arnaud Guillin. On the rate of convergence in Wasserstein distance of the empirical measure. *Probability Theory and Related Fields*, 162 (3–4):707–738, 2015. doi: 10.1007/s00440-014-0583-7.

João Gama, Pedro Medas, Gladys Castillo, and Pedro Rodrigues. Learning with drift detection. In *Advances in Artificial Intelligence — SBIA 2004*, volume 3171 of *Lecture Notes in Computer Science*, pages 286–295. Springer, 2004. doi: 10.1007/978-3-540-28645-5_29.

João Gama, Indrė Žliobaitė, Albert Bifet, Mykola Pechenizkiy, and Abdelhamid Bouchachia. A survey on concept drift adaptation. *ACM Computing Surveys*, 46 (4):44:1–44:37, 2014. doi: 10.1145/2523813.

Aude Genevay, Lénaïc Chizat, Francis Bach, Marco Cuturi, and Gabriel Peyré. Sample complexity of Sinkhorn divergences. In *Proceedings of the 22nd International Conference on Artificial Intelligence and Statistics (AISTATS)*, volume 89 of *Proceedings of Machine Learning Research*, pages 1574–1583, 2019.

Ziv Goldfeld, Kengo Kato, Gabriel Rioux, and Ritwik Sadhu. Limit theorems for entropic optimal transport maps and the sinkhorn divergence. *Electronic Journal of Statistics*, 18(1):980–1041, 2024. doi: 10.1214/24-EJS2217.

Fabian Hinder, Valerie Vaquet, and Barbara Hammer. The window dilemma: Why concept drift detection is ill-posed, 2023. Preprint.

Andrew H. Jazwinski. *Stochastic Processes and Filtering Theory*, volume 64 of *Mathematics in Science and Engineering*. Academic Press, New York, 1970.

Jiashuo Jiang, Xiaocheng Li, and Jiawei Zhang. Online stochastic optimization with wasserstein based non-stationarity, 2020. URL https://arxiv.org/abs/2012.06961.

Rudolf E. Kalman. A new approach to linear filtering and prediction problems. *Journal of Basic Engineering*, 82(1):35–45, 1960. doi: 10.1115/1.3662552.

Sanjit Kaul, Roy Yates, and Marco Gruteser. Real-time status: How often should one update? In *Proceedings of IEEE INFOCOM*, pages 2731–2735, 2012. doi: 10.1109/INFCOM.2012.6195689.

Dominic S. T. Keehan, Edward J. Anderson, and Wolfram Wiesemann. Don't look back in anger: Wasserstein distributionally robust optimization with nonstationary data. *CoRR*, abs/2510.18566, 2025. doi: 10.48550/arXiv.2510.18566.

Alessio Mazzetto and Eli Upfal. Nonparametric density estimation under distribution drift. In *Proceedings of the 40th International Conference on Machine Learning*, volume 202 of *Proceedings of Machine Learning Research*, pages 24251–24270. PMLR, 2023. URL https://proceedings.mlr.press/v202/mazzetto23a.html.

Jonathan Niles-Weed and Francis Bach. Sharp asymptotic and finite-sample rates of convergence of empirical measures in Wasserstein distance. *Bernoulli*, 25(4A): 2620–2648, 2019. doi: 10.3150/18-BEJ1077.

Zhiyong Wang, Jize Xie, Yi Chen, John C. S. Lui, and Dongruo Zhou. Variance-dependent regret bounds for non-stationary linear bandits, 2024. URL https://arxiv.org/abs/2403.10732.

Roy D. Yates, Yin Sun, D. Richard Brown, Sanjit K. Kaul, Eytan Modiano, and Sennur Ulukus. Age of information: An introduction and survey. *IEEE Journal on Selected Areas in Communications*, 39(5):1183–1210, 2021. doi: 10.1109/JSAC.2021.3064980.

Jianjun Yuan and Andrew G. Lamperski. Trading-off static and dynamic regret in online least-squares and beyond. In *The Thirty-Fourth AAAI Conference on Artificial Intelligence*, pages 6712–6719. AAAI Press, 2020. doi: 10.1609/AAAI.V34I04.6149.